

BUSINESS AI / ML • CASE STUDY

AI-Powered Fraud Detection in Digital Payments

A Case Study of PhonePe

DATA

Transactions



AI / ML

Patterns



ACTION

Risk response

BBA FinTech | Chitkara University

SUBMITTED BY

Team Members

Angel

2520992529

Ayushi

2520992558

Arpita

2520992543

Arshita

2520992546

Ankita

2520992533

Aniketa

2520992531

Anchit

2520992527

AI • SECURITY • DIGITAL PAYMENTS

Company & Industry Overview

PhonePe: India's Digital Payments Platform

PhonePe is an Indian fintech and digital payments platform offering UPI payments, bill payments, merchant payments and access to financial services. It operates in a high-volume, real-time digital payments environment where trust and security are critical.

Industry Context

FinTech / Digital Payments

- Mobile-first transactions
- Real-time processing
- High fraud exposure
- Strong need for security

71.8 Cr

Registered users

5 Cr+

Merchants accepting PhonePe

Guardian

Risk & fraud-detection platform

Real-Time Fraud Detection Without Friction

How can PhonePe identify and prevent potentially fraudulent transactions in real time — while keeping genuine payments convenient?

Why the problem matters

Digital payment fraud can cause financial loss, customer dissatisfaction and reduced trust. The system must make fast decisions while handling very large transaction volumes.

Fraud patterns to detect

- Social engineering & phishing
- Identity theft
- Fraudulent accounts
- Suspicious beneficiaries
- Unauthorized transactions
- Malicious apps / devices

Core business challenge

Balance security with convenience:

Too strict → genuine users are blocked.
Too loose → fraudulent activity may pass through.

Data Needed for AI/ML Fraud Detection

Transaction data

Amount, date/time, sender, receiver, frequency, payment instrument, status and history

Behavioural data

Typical amounts, transaction frequency, login behaviour and unusual activity changes

Device data

Device type/OS, settings, fingerprint and new-device login signals

Location data

Transaction location, location changes, geo-fencing and location-risk signals

Account & relationship

Phone numbers, user IDs, bank accounts, VPAs and merchant relationships

Historical fraud data

Reported fraud accounts, transactions, complaints and known fraud patterns

Proposed AI-Powered Risk Assessment

A multi-technique approach

A practical fraud-detection system should combine multiple signals and AI/ML techniques rather than depend on one model.

The case study uses a conceptual architecture based on PhonePe's publicly described risk scoring, behavioural signals, device intelligence and relationship analysis.

Classification

Categorises transactions by risk level

Anomaly detection

Flags activity that differs from normal behaviour

Risk scoring

Combines signals into a transaction risk assessment

Graph analysis

Maps relationships across users, accounts and devices

Behavioural analysis

Learns a user's typical activity pattern

How the AI Solution Works



Expected / Existing Business Benefits

Reduced fraud

Early identification can reduce fraudulent transactions and associated financial loss.

Faster decisions

AI/ML can analyse large volumes of signals quickly for real-time decisions.

Customer Protection

Risk warnings and interventions can help users recognise suspicious payments.

Improved trust

A safer payment environment supports confidence in digital payments.

Less manual work

Automation lets human teams focus on complex or high-risk cases.

Better fraud intelligence

Data analysis helps identify changing and emerging fraud patterns.

Scale indicator

PhonePe's FY2023–24 Annual Report states that its Guardian platform performed 3,000+ crore evaluations daily.

Risks, Ethics & Responsible AI

Privacy

Use only appropriate data with strong security, access controls and permission.

False positives

A genuine transaction may be flagged, creating friction or inconvenience.

False negatives

Fraud may be missed, creating financial loss and customer dissatisfaction.

Bias & fairness

Model performance can differ across groups; evaluation is required.

Transparency

Users need understandable explanations when a payment is flagged or stopped.

Human oversight

Complex/high-risk cases should remain reviewable by people.

Evolving threat

Fraudsters adapt through phishing, deepfakes and misuse of generative AI, so controls and models must be continuously evaluated.

Tools and Prompts Used

ChatGPT

- Research brainstorming
- Simplifying AI/ML concepts
- Structuring the case study
- Improving slide wording

Web / Company Sources

- PhonePe Security & Trust & Safety material
- FY2023–24 Annual Report
- Public information used to ground claims

PowerPoint / AI-assisted design

- Slide organization
- Process-flow visualization
- Converting research into presentation-ready points

Important prompts used

1. “Explain how AI/ML can detect fraud in digital payments using PhonePe as a case study.”
2. “Suggest the data required for real-time digital payment fraud detection.”
3. “Create a simple flowchart: transaction → data analysis → risk assessment → decision → feedback.”
4. “List responsible-AI risks: privacy, bias, false positives, transparency and human oversight.”

Prompt Improvement & AI Evaluation

Original / Basic Prompt

“Tell me about AI fraud detection in PhonePe.”

Improved Prompt

“For a BBA FinTech & AI case study on PhonePe, explain real-time fraud detection in 5 parts: data required, ML techniques, process flow, business impact and responsible-AI risks. Keep it concise and presentation-ready.”

Difference in output

- More structured
- More relevant
- Less generic
- Easier to convert into slides

What I learned

- Context improves relevance
- Output format reduces rewriting
- Constraints control length and level

AI evaluation

Checked for:

- Relevance & clarity
- Accuracy against sources
- No invented proprietary details

Recommendations & Conclusion

Recommendations

1. Real-Time Monitoring

Strengthen monitoring for decisions needed within seconds.

2. Multi-Method Detection

Combine classification, anomaly, behavioural, graph analysis and risk scoring.

3. Reduce False Positives

Avoid unnecessarily blocking unusual but legitimate behaviour.

4. Improve Explainability

Provide simple reasons when payments are flagged or stopped.

5. Human Oversight

Keep human review for complex/high-risk cases and continuously update models

6. Privacy & Awareness

Strengthen privacy controls and user awareness against social-engineering scams.

Case conclusion

AI/ML-Based Fraud Detection

AI/ML can help PhonePe move from rule-only detection toward a multi-signal, real-time risk assessment approach.

Balanced Protection

The objective is not simply to block fraud, but to protect users while preserving legitimate payment experiences.

Key takeaway : Fraud Detection Cycle

1. Detect → Identify suspicious or unusual transactions.

2. Assess → Evaluate the level of fraud risk using multiple signals.

3. Act → Block, flag, or allow the transaction based on the risk.

4. Learn → Use new fraud patterns and outcomes to improve detection.

5. Improve → Continuously update models and rules for better accuracy.

Detect → Assess → Act → Learn → Improve

Learning Reflection, References & GitHub

Learning Reflection

1. How transaction, behavioural, device, location and historical data can be combined for fraud detection.
2. Applying AI/ML to a real business problem requires balancing detection accuracy with customer convenience.
3. I learned that prompt engineering makes AI-assisted research more focused, structured and presentation-ready.

Example:

Verify AI-Generated Claims

AI-generated claims had to be checked against PhonePe's public security material and annual report rather than treated as automatically correct.

Refine Prompts

Adding context, required sections and length constraints reduced generic output and rewriting.

Individual GitHub Repository

Anchit Phutela
<https://github.com/anchitphutela-max>

References

1. PhonePe — PhonePe Security: Every Payment is Protected on PhonePe.
2. PhonePe — Guardrails: Real-Time Fraud Detection (11 Feb 2025).
3. PhonePe — Guardrails: Advanced Risk Detection Features (27 Mar 2025).
4. PhonePe — Guardrails: Foundations of Trust & Safety (23 Jan 2025).
5. PhonePe — PhonePe Protect (3 Nov 2025).
6. PhonePe Annual Report FY2023–24

Thank You

A Case Study of PhonePe

DATA

Transactions



AI / ML

Patterns



ACTION

Risk response

BBA FinTech | Chitkara University

SUBMITTED BY

Team Members

Angel

<https://github.com/Angelrawal1207>

Ayushi

<https://github.com/ayushi27-08>

Arpita

<https://github.com/arpitaarora1>

Arshita

<https://github.com/arshitakumar3110>

Ankita

<https://github.com/ankitachauhan12>

Aniketa

<https://github.com/aniketalohan-creator>

Anchit

<https://github.com/anchitphutela-max>